

The Crystal Realm of LUMINARA

Cisco Packet Tracer Implementation | Student ID: 22201501 | Base Network: 15.1.0.0/16

Item	Final Value
Student ID used	22201501
Last four digits	1501
Base network	15.1.0.0/16
Group size	2 members
Web domain	www.luminara.net
Email domain	luminara.net
Required homepage message	Welcome to the Crystal Realm of LUMINARA

1. Project Overview

This report documents the final working Cisco Packet Tracer implementation for **The Crystal Realm of LUMINARA**. The network contains six routed LANs, VLSM addressing based on student ID 22201501, RIP v2 dynamic routing, required static routes, DHCP relay for Frostfield Ranch and Aurora Ridge, and Photon Cove services for Web, DNS, Email, and DHCP.

2. Work Distribution for 2 Members

Group Member	Assigned Work
Member 1: Hosni Rabbani	VLSM calculation, base network derivation, IP address planning, network notes/labels, and report preparation.
Member 2: Shahrin Tabassum	Packet Tracer topology, cabling, router interface configuration, RIP v2/static routing, server setup, DHCP/DNS/Web/Email testing.

3. VLSM Subnetting Plan

Student ID 22201501 gives last four digits 1501, split as 15 and 01. Therefore, the base address is 15.1.0.0/16.

Location	Population	Subnet	Subnet Mask	Usable Hosts	Gateway
Granite Borough	5200	15.1.0.0/19	255.255.224.0	8190	15.1.0.1
Corsair Isles	1200	15.1.32.0/21	255.255.248.0	2046	15.1.32.1
Frostfield Ranch	350	15.1.40.0/23	255.255.254.0	510	15.1.40.1
Aurora Ridge	150	15.1.42.0/24	255.255.255.0	254	15.1.42.1
Photon Cove	80	15.1.43.0/25	255.255.255.128	126	15.1.43.1
Sunspire Palace	45	15.1.43.128/26	255.255.255.192	62	15.1.43.129

4. VLSM Tree

VLSM Tree

Base Network: 15.1.0.0/16

```
15.1.0.0/16
|
|-- 15.1.0.0/19      Granite Borough
|-- 15.1.32.0/21     Corsair Isles
|-- 15.1.40.0/23     Frostfield Ranch
|-- 15.1.42.0/24     Aurora Ridge
|-- 15.1.43.0/25     Photon Cove
|-- 15.1.43.128/26   Sunspire Palace
|
|-- 15.1.43.192/30   R-SP <-> R-PC WAN
|-- 15.1.43.196/30   R-PC <-> R-GB WAN
|-- 15.1.43.200/30   R-SP <-> R-AR WAN
|-- 15.1.43.204/30   R-AR <-> R-GB WAN
|-- 15.1.43.208/30   R-GB <-> R-CI WAN
|-- 15.1.43.212/30   R-PC <-> R-FR WAN
```

5. WAN Addressing Table

Link	Network	Side A	Side A IP	Side B	Side B IP
R-SP <-> R-PC	15.1.43.192/30	R-SP S0/1/0	15.1.43.193	R-PC S0/1/0	15.1.43.194
R-PC <-> R-GB	15.1.43.196/30	R-PC S0/0/1	15.1.43.197	R-GB S0/0/0	15.1.43.198
R-SP <-> R-AR	15.1.43.200/30	R-SP S0/1/1	15.1.43.201	R-AR S0/1/1	15.1.43.202
R-AR <-> R-GB	15.1.43.204/30	R-AR S0/1/0	15.1.43.205	R-GB S0/0/1	15.1.43.206
R-GB <-> R-CI	15.1.43.208/30	R-GB S0/1/1	15.1.43.209	R-CI S0/1/0	15.1.43.210
R-PC <-> R-FR	15.1.43.212/30	R-PC S0/0/0	15.1.43.213	R-FR S0/1/0	15.1.43.214

6. Server and End Device Addressing

Device	Location	IP Address	Subnet Mask	Gateway	DNS
Web Server	Photon Cove	15.1.43.10	255.255.255.128	15.1.43.1	15.1.43.11
DNS Server	Photon Cove	15.1.43.11	255.255.255.128	15.1.43.1	15.1.43.11
Email Server	Photon Cove	15.1.43.12	255.255.255.128	15.1.43.1	15.1.43.11
DHCP Server	Photon Cove	15.1.43.13	255.255.255.128	15.1.43.1	15.1.43.11
SP-PC1	Sunspire Palace	15.1.43.130	255.255.255.192	15.1.43.129	15.1.43.11
SP-PC2	Sunspire Palace	15.1.43.131	255.255.255.192	15.1.43.129	15.1.43.11
GB-PC1	Granite Borough	15.1.0.2	255.255.224.0	15.1.0.1	15.1.43.11
GB-PC2	Granite Borough	15.1.0.3	255.255.224.0	15.1.0.1	15.1.43.11
CI-PC1	Corsair Isles	15.1.32.2	255.255.248.0	15.1.32.1	15.1.43.11
CI-PC2	Corsair Isles	15.1.32.3	255.255.248.0	15.1.32.1	15.1.43.11
FR-PC1/FR-PC2	Frostfield Ranch	DHCP: 15.1.40.x	255.255.254.0	15.1.40.1	15.1.43.11
AR-PC1/AR-PC2	Aurora Ridge	DHCP: 15.1.42.x	255.255.255.0	15.1.42.1	15.1.43.11

7. Server Services Configuration

DHCP Server Pools

```
DHCP Server IP: 15.1.43.13/25
Default Gateway: 15.1.43.1
DNS Server: 15.1.43.11
```

```
Pool: FROSTFIELD
Default Gateway: 15.1.40.1
DNS Server: 15.1.43.11
Start IP Address: 15.1.40.10
Subnet Mask: 255.255.254.0
```

```
Pool: AURORA
Default Gateway: 15.1.42.1
DNS Server: 15.1.43.11
Start IP Address: 15.1.42.10
Subnet Mask: 255.255.255.0
```

DNS, Web, and Email Services

```
DNS Records:
www.luminara.net -> 15.1.43.10
mail.luminara.net -> 15.1.43.12
```

```
Web Homepage Message:
Welcome to the Crystal Realm of LUMINARA
```

```
Email Domain: luminara.net
Email Users: sun / 123, photon / 123, granite / 123
```

8. Routing Strategy

Static routing and RIP v2: The final configuration uses RIP v2 with no auto-summary on all routers. Required static routes are configured for Sunspire Palace to Photon Cove, Photon Cove to Granite Borough, and Sunspire Palace to Aurora Ridge. A primary static route from Sunspire Palace to Granite Borough uses Photon Cove, while a floating static route through Aurora Ridge uses administrative distance 130. No 0.0.0.0/0 default route is used.

9. Verification Screenshots

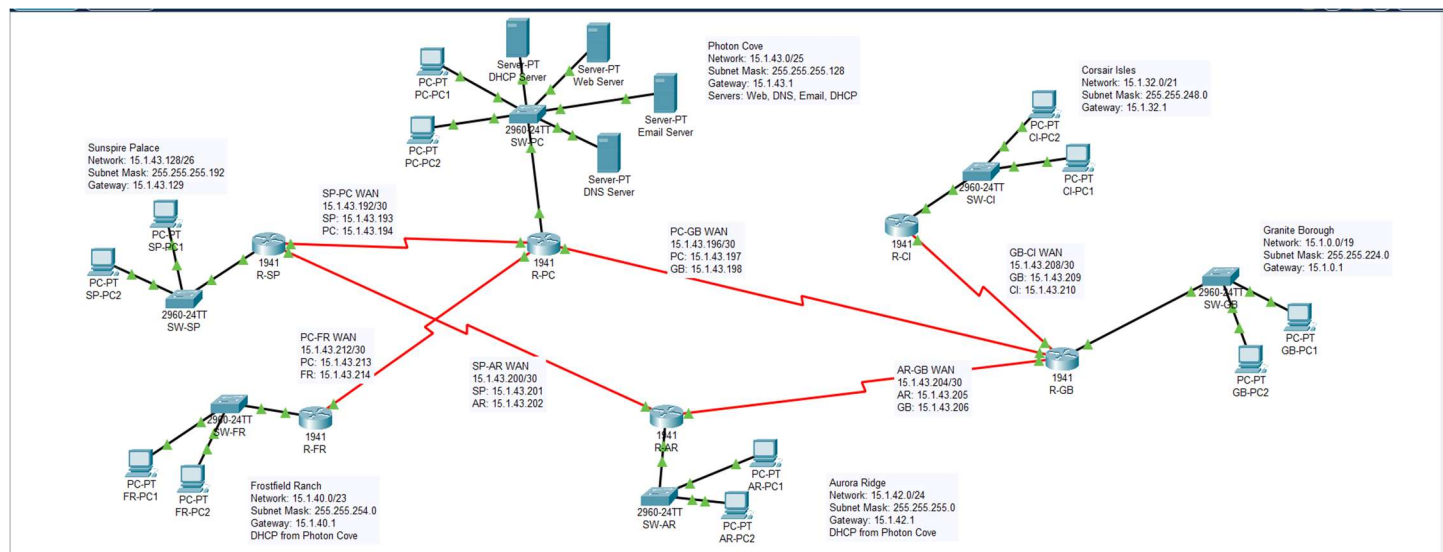


Figure 1: Final Topology Diagram with Network Notes

Command Prompt

```
C:\>ping 15.1.43.1

Pinging 15.1.43.1 with 32 bytes of data:

Reply from 15.1.43.1: bytes=32 time=3ms TTL=254
Reply from 15.1.43.1: bytes=32 time=1ms TTL=254
Reply from 15.1.43.1: bytes=32 time=3ms TTL=254
Reply from 15.1.43.1: bytes=32 time=1ms TTL=254

Ping statistics for 15.1.43.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\>ping 15.1.0.1

Pinging 15.1.0.1 with 32 bytes of data:

Reply from 15.1.0.1: bytes=32 time=11ms TTL=253
Reply from 15.1.0.1: bytes=32 time=4ms TTL=253
Reply from 15.1.0.1: bytes=32 time=10ms TTL=253
Reply from 15.1.0.1: bytes=32 time=2ms TTL=253

Ping statistics for 15.1.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 11ms, Average = 6ms

C:\>ping 15.1.40.1

Pinging 15.1.40.1 with 32 bytes of data:

Reply from 15.1.40.1: bytes=32 time=2ms TTL=253
Reply from 15.1.40.1: bytes=32 time=5ms TTL=253
Reply from 15.1.40.1: bytes=32 time=2ms TTL=253
Reply from 15.1.40.1: bytes=32 time=2ms TTL=253

Ping statistics for 15.1.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 5ms, Average = 2ms

C:\>ping 15.1.32.1

Pinging 15.1.32.1 with 32 bytes of data:

Reply from 15.1.32.1: bytes=32 time=8ms TTL=252
Reply from 15.1.32.1: bytes=32 time=3ms TTL=252
Reply from 15.1.32.1: bytes=32 time=6ms TTL=252
Reply from 15.1.32.1: bytes=32 time=4ms TTL=252

Ping statistics for 15.1.32.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

Figure 2: Ping Verification - Part 1

```
Ping statistics for 15.1.0.1:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 2ms, Maximum = 11ms, Average = 6ms
```

```
C:\>ping 15.1.40.1
```

```
Pinging 15.1.40.1 with 32 bytes of data:
```

```
Reply from 15.1.40.1: bytes=32 time=2ms TTL=253  
Reply from 15.1.40.1: bytes=32 time=5ms TTL=253  
Reply from 15.1.40.1: bytes=32 time=2ms TTL=253  
Reply from 15.1.40.1: bytes=32 time=2ms TTL=253
```

```
Ping statistics for 15.1.40.1:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 2ms, Maximum = 5ms, Average = 2ms
```

```
C:\>ping 15.1.32.1
```

```
Pinging 15.1.32.1 with 32 bytes of data:
```

```
Reply from 15.1.32.1: bytes=32 time=8ms TTL=252  
Reply from 15.1.32.1: bytes=32 time=3ms TTL=252  
Reply from 15.1.32.1: bytes=32 time=6ms TTL=252  
Reply from 15.1.32.1: bytes=32 time=4ms TTL=252
```

```
Ping statistics for 15.1.32.1:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 3ms, Maximum = 8ms, Average = 5ms
```

```
C:\>ping 15.1.42.1
```

```
Pinging 15.1.42.1 with 32 bytes of data:
```

```
Reply from 15.1.42.1: bytes=32 time=8ms TTL=254  
Reply from 15.1.42.1: bytes=32 time=2ms TTL=254  
Reply from 15.1.42.1: bytes=32 time=4ms TTL=254  
Reply from 15.1.42.1: bytes=32 time=1ms TTL=254
```

```
Ping statistics for 15.1.42.1:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 1ms, Maximum = 8ms, Average = 3ms
```

Figure 3: Ping Verification - Part 2

AR-PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 15.1.42.10

Subnet Mask 255.255.255.0

Default Gateway 15.1.42.1

DNS Server 15.1.43.11

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::209:7CFF:FE25:A2D9

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

Figure 4: Aurora Ridge DHCP Verification

FR-PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 15.1.40.10

Subnet Mask 255.255.254.0

Default Gateway 15.1.40.1

DNS Server 15.1.43.11

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:43FF:FE1C:4A57

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Top

Figure 5: Frostfield Ranch DHCP Verification

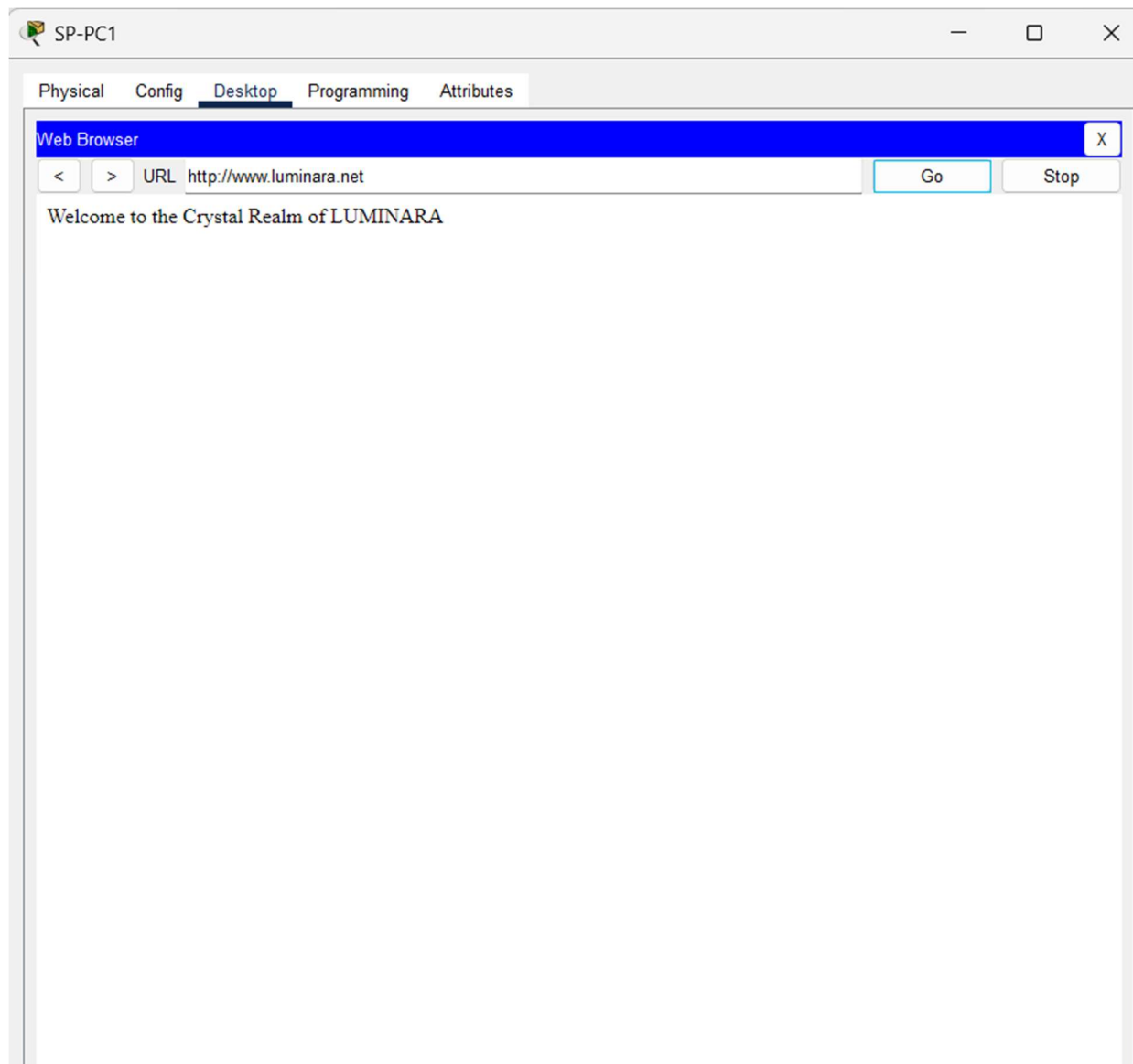


Figure 6: Web Access Verification

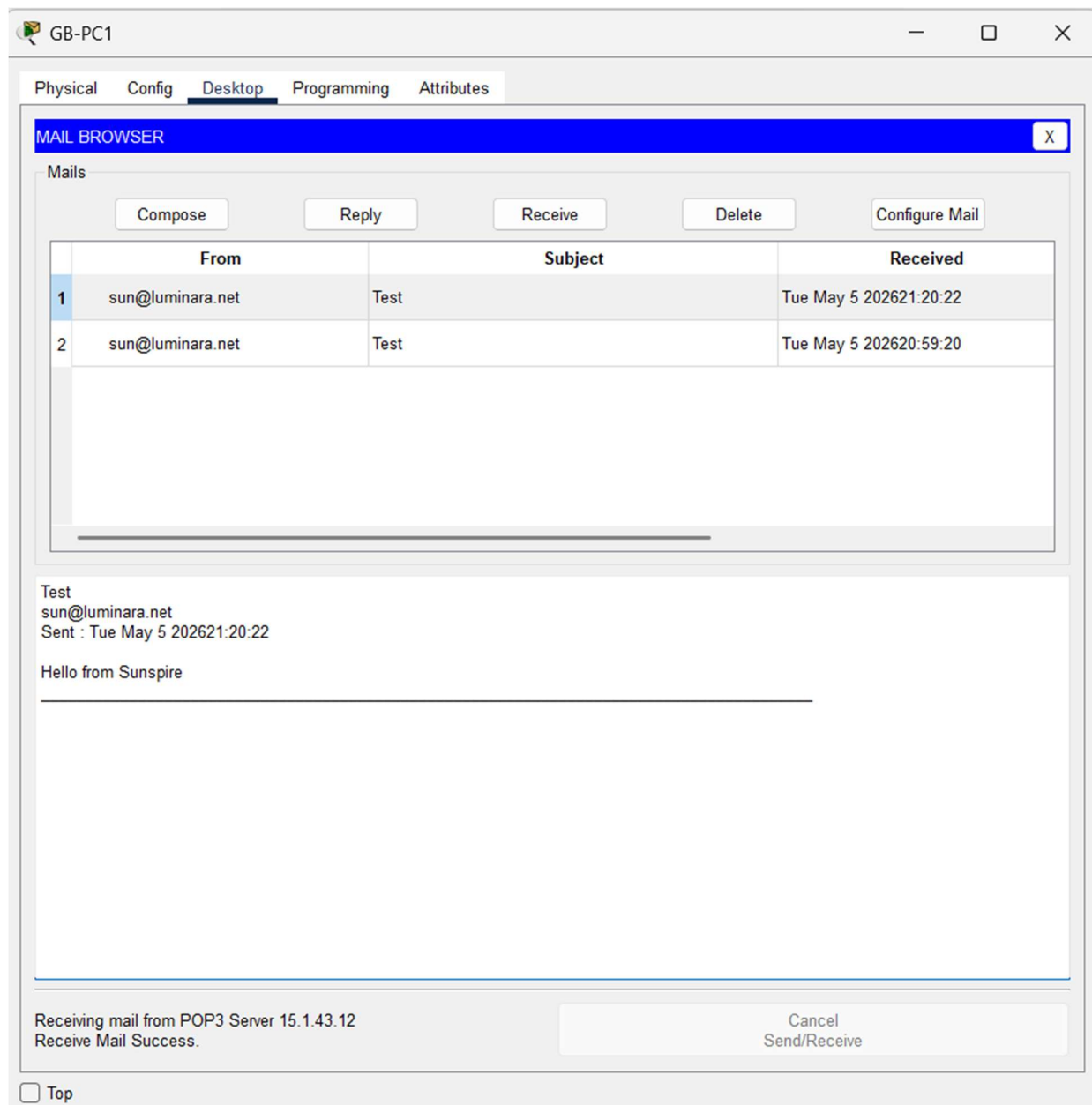


Figure 7: Email Service Verification

10. Final Verification Checklist

Verification Item	Status
Separate routed LANs	Implemented for all six locations.
VLSM subnetting	Completed from base network 15.1.0.0/16.
Router interfaces	All required LAN/WAN interfaces configured and verified up/up.
RIP v2	Configured on all routers with no auto-summary.
Static/floating routes	Primary route to Granite via Photon Cove and floating route via Aurora Ridge configured.
No default route	No 0.0.0.0/0 default route configured.
DHCP	Frostfield Ranch and Aurora Ridge received IP addresses from Photon Cove DHCP Server.
Web/DNS	www.luminara.net opened with required homepage message.
Email	Email received successfully between users in different LANs.

11. Final Router Running Configurations

The following commands are copied from the final working Packet Tracer running-config outputs after the successful ping, DHCP, web, and email tests.

R-SP Final Running Configuration

```
R-SP:
R-SP>en
R-SP#show running-config
Building configuration...

Current configuration : 1034 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R-SP
!
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
license udi pid CISCO1941/K9 sn FTX15244216-
!
!
!
!
!
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
interface GigabitEthernet0/0
ip address 15.1.43.129 255.255.255.192
duplex auto
speed auto
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/1/0
ip address 15.1.43.193 255.255.255.252
clock rate 64000
!
interface Serial0/1/1
ip address 15.1.43.201 255.255.255.252
clock rate 64000
!
interface Vlan1
no ip address
shutdown
!
router rip
version 2
network 15.0.0.0
no auto-summary
!
ip classless
ip route 15.1.43.0 255.255.255.128 15.1.43.194
ip route 15.1.42.0 255.255.255.0 15.1.43.202
ip route 15.1.0.0 255.255.224.0 15.1.43.194
ip route 15.1.0.0 255.255.224.0 15.1.43.202 130
!
ip flow-export version 9
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
end
```

R-PC Final Running Configuration

```
R-PC:

R-PC#show running-config
Building configuration...

Current configuration : 1095 bytes
!
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R-PC
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO1941/K9 sn FTX1524CYXQ-
!
!
!
!
!
!
!
no ip domain-lookup
!
!
spanning-tree mode pvst
!
!
!
!
!
interface GigabitEthernet0/0
ip address 15.1.43.1 255.255.255.128
duplex auto
speed auto
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/0/0
ip address 15.1.43.213 255.255.255.252
clock rate 64000
!
interface Serial0/0/1
ip address 15.1.43.197 255.255.255.252
clock rate 2000000
!
interface Serial0/1/0
ip address 15.1.43.194 255.255.255.252
!
interface Serial0/1/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
router rip
version 2
network 15.0.0.0
no auto-summary
!
ip classless
ip route 15.1.0.0 255.255.224.0 15.1.43.198
ip route 15.1.43.128 255.255.255.192 15.1.43.193
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end
```

R-FR Final Running Configuration

```
R-FR:

R-FR#show running-config
Building configuration...

Current configuration : 862 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R-FR
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO1941/K9 sn FTX15240R08-
!
!
!
!
!
!
!
!
no ip domain-lookup
!
!
spanning-tree mode pvst
!
!
!
!
!
interface GigabitEthernet0/0
ip address 15.1.40.1 255.255.254.0
ip helper-address 15.1.43.13
duplex auto
speed auto
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/1/0
ip address 15.1.43.214 255.255.255.252
!
interface Serial0/1/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
router rip
version 2
network 15.0.0.0
no auto-summary
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end
```

R-CI Final Running Configuration

```
R-CI:

R-CI>en
R-CI#show running-config
Building configuration...

Current configuration : 832 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
```

```

hostname R-CI
!
!
!
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO1941/K9 sn FTX1524ET59-
!
!
!
!
!
!
!
!
!
!
no ip domain-lookup
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
ip address 15.1.32.1 255.255.248.0
duplex auto
speed auto
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/1/0
ip address 15.1.43.210 255.255.255.252
!
interface Serial0/1/1
no ip address
clock rate 2000000
shutdown
!
interface Vlan1
no ip address
shutdown
!
router rip
version 2
network 15.0.0.0
no auto-summary
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
!
end

```

R-GB Final Running Configuration

```

R-GB:

R-GB#show running-config
Building configuration...

Current configuration : 1125 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R-GB
!
!
!
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO1941/K9 sn FTX152483QA-
!
!

```

```

!
!
!
!
!
!
!
no ip domain-lookup
!
!
spanning-tree mode pvst
!
!
!
!
!
!
!
interface GigabitEthernet0/0
ip address 15.1.0.1 255.255.224.0
duplex auto
speed auto
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/0/0
ip address 15.1.43.198 255.255.255.252
!
interface Serial0/0/1
ip address 15.1.43.206 255.255.255.252
!
interface Serial0/1/0
no ip address
clock rate 2000000
shutdown
!
interface Serial0/1/1
ip address 15.1.43.209 255.255.255.252
clock rate 64000
!
interface Vlan1
no ip address
shutdown
!
router rip
version 2
network 15.0.0.0
no auto-summary
!
ip classless
ip route 15.1.43.128 255.255.255.192 15.1.43.197
ip route 15.1.43.192 255.255.255.252 15.1.43.197
ip route 15.1.43.0 255.255.255.128 15.1.43.197
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end

```

R-AR Final Running Configuration

```
R-AR:
R-AR>en
R-AR#show running-config
Building configuration...

Current configuration : 925 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R-AR
!
!
!
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO1941/K9 sn FTX1524M1US-
!
!
!
!
!
!
!
```



```
no ip domain-lookup
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
ip address 15.1.42.1 255.255.255.0
ip helper-address 15.1.43.13
duplex auto
speed auto
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/1/0
ip address 15.1.43.205 255.255.255.252
clock rate 64000
!
interface Serial0/1/1
ip address 15.1.43.202 255.255.255.252
!
interface Vlan1
no ip address
shutdown
!
router rip
version 2
network 15.0.0.0
no auto-summary
!
!
ip classless
ip route 15.1.43.128 255.255.255.192 15.1.43.201
!
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end
```